January 1, 1918

Date of Revision

INJECTORS

No new injectors of the lifting type or back bodies for lifting injectors are to be purchased for use on locomotives.

All new injectors purchased will be of the non-lifting type.

In order to gradually work out the lifting type, certain classes are to receive non-lifting injectors. It is to be understood that serviceable lifting injectors are not to be removed except to provide a sufficient stock to take care of repairs to engines not authorized to receive non-lifting injectors. Since the change to non-lifting injector can only be made on engines held out of service for repairs, the requirements of lifting injectors for stock should be studied whenever an engine authorized to receive non-lifting injectors is held out of service for repairs.

The table on page 128-B shows the authorized type and size of injector for each locomotive class. This table shows the proper size of non-lifting injector which is to replace the lifting type in certain classes. The first non-lifting injector applied to a locomotive should be applied to the left side.

When the back body of a lifting type of injector is no longer serviceable, the injector should be dismantled, serviceable parts returned to stock and parts unfit for service boxed up and shipped to the storekeeper at Topeka.

Since no engines using size 9-A injectors are authorized to receive non-lifting injectors, it is possible a shortage of this size may occur. In this case a 10-1/2-A should be applied.

Classes using 10-1/2-A injectors may have 11-A applied in case 10-1/2-A are not available. Such classes should be equipped with 11-A in order to provide 10-1/2-A for use on smaller engines.

No new 13-M injectors or 2-1/2"-M starting valves are to be purchased. When complete renewal is necessary and none can be obtained from stock, T-60 injectors and 2-1/2"-T starting valves are to be applied. Likewise, 2"-T starting valves to be used for renewal of 2"-R.

APPLICATION OF INJECTORS AND INJECTOR PIPING SYSTEM

Diagrams showing the arrangement of the piping system in a general way and referring to the fittings in detail are shown by pages 129-C and 129-D for lifting and non-lifting injectors respectively. The equipment on all locomotives may not agree with the details shown. Future renewals and applications of new equipment are to be made in accordance with these pages.

It is important that injector and injector piping joints shall be secured properly and safely. As locomotives pass through shops or when injector piping equipment is repaired in roundhouses, careful inspection must be made to see that pipes are properly secured to sleeves, that the shoulders on sleeves and nuts are large enough and in good condition to hold, that threads are in good condition, that nuts are not cracked and not stretched so as to be too large for corresponding thread and that brazing is in good condition. The proper method of brazing is shown by page 158 and the proper forms of sleeves and nuts are shown by pages 129-E, 129-F, 129-G and 129-H.

To allow for expansion and contraction, injector steam pipes should be bent with easy curves and the bends should not be abrupt. The actual length of pipe should be much greater than that of a straight line between terminal connections. The bends in the pipe should be on a radius of at least 10" to 14". Pockets in which water might collect should be avoided.

TESTING INJECTORS:

Injectors must be tested before each trip or day's work. The rules given below are to be followed in testing each injector before an engine is turned over to its crew:

(a) See that steam pressure in boiler is not less than 125 lbs.
(b) See that water level in boiler is not less than 3" nor more than 6" above lowest reading of water glass.
(c) See that main steam fountain valve to injectors is wide open.
(d) See that tank valve is wide open.
(e) Open overflow valve wide open if injector is of the non-lifting type, or see that overflow valve is wide open if injector is of the lifting type.
(f) Open injector water valve wide open.
(g) Prime injector, then pull steam rum wide open and note whether injector delivers water into the boiler satisfactorily.

If injector is equipped with telltale device for closed overflow operation, the following test is to be made in addition to that outlined above:

(h) With injector delivering water to boiler, close overflow valve tight.
(i) Cause telltale device to operate by gradually closing injector water valve until delivery of water to boiler is interrupted, at which time steam should begin blowing from the telltale nozzle.

Apply following test to water and overflow valves of each injector (lifting or non-lifting):

(a) Close water valve tight.
(b) Close overflow valve tight.
(c) Open steam rum full and note whether steam appears at overflow or blows back through tank hose into tender. Steam should blow continuously from telltale device of a closed overflow injector while this test is being made.

Locomotives are not to be dispatched from any terminal unless both units of water supply to boiler are in safe and suitable condition for service.

Rewritten 6-24-48, account other page torn.